FLATNESS AFTER A BASE CHANGE MAKES THE NATURAL MAP ON TOR ONE SURJECTIVE

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Allowed standard prerequisites (editorial).

- Tor computed from a free resolution. Used at the start of the counted proof (lines 24072-24076). Identifies the two Tor groups as quotients of the cycles at the second step of the resolution.
- Flatness as preservation of exactness. Used at the middle of the counted proof (lines 24084-24091). Keeps the exact sequence exact after tensoring the second base change, which yields surjectivity of the map on cycles.

Lemma 0.1. *Let $R \to R' \to R''$ be ring maps. Let M be an R-module. Suppose that $M \otimes_R R'$ is flat over R'. Then the natural map $Tor_1^R(M, R') \otimes_{R'} R'' \to Tor_1^R(M, R'')$ is onto.*

Proof. Let $F_\bullet$ be a free resolution of M over R. The complex $F_2 \otimes_R R' \to F_1 \otimes_R R' \to F_0 \otimes_R R'$ computes $\mathrm{Tor}_1^R(M, R')$. The complex $F_2 \otimes_R R'' \to F_1 \otimes_R R'' \to F_0 \otimes_R R''$ computes $\mathrm{Tor}_1^R(M, R'')$. Note that $F_i \otimes_R R' \otimes_{R'} R'' = F_i \otimes_R R''$. Let $K' = \mathrm{Ker}(F_1 \otimes_R R' \to F_0 \otimes_R R')$ and similarly $K'' = \mathrm{Ker}(F_1 \otimes_R R'' \to F_0 \otimes_R R'')$. Thus we have an exact sequence

$$0 \to K' \to F_1 \otimes_R R' \to F_0 \otimes_R R' \to M \otimes_R R' \to 0.$$

By the assumption that $M \otimes_R R'$ is flat over R', the sequence

$$K' \otimes_{R'} R'' \to F_1 \otimes_R R'' \to F_0 \otimes_R R'' \to M \otimes_R R'' \to 0$$

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is still exact. This means that $K' \otimes_{R'} R'' \to K''$ is surjective. Since $\mathrm{Tor}_1^R(M, R')$ is a quotient of K' and $\mathrm{Tor}_1^R(M, R'')$ is a quotient of K'' we win. $\square$